

# Shahab Derhami

Postdoctoral Fellow  
School of Industrial & Systems Engineering  
Georgia Institute of Technology

755 Ferst Dr NW, Atlanta, GA  
shahab.derhami@isye.gatech.edu  
www.shahabderhami.com

---

## Education

Ph.D. Industrial & Systems Engineering, Auburn University, AL, 2017 (GPA: 4.0/4.0)  
M.E. Industrial & Systems Engineering, Auburn University, AL, 2012 (GPA: 4.0/4.0)  
M.S. Social & Economic Systems Engineering, Bu-Ali Sina University, Iran, 2006  
B.S. Industrial Engineering, Azad University, South Tehran branch, Iran, 2004

## Employment

**Postdoctoral Fellow**, Georgia Institute of Technology, GA, 2017-Present  
**Instructor, Research Assistant, and Teaching Assistant**, Auburn University, AL, 2011-2017  
**Economic Analyst**, Part Lastic Factories Group, Iran, 2008-2011  
**Production & Project Planning Specialist**, Atlas Afrooz Shargh Company, Iran, 2006-2008  
**Instructor**, Department of Industrial Engineering, Azad University, Iran, 2004-2005

## Publications

### Journal papers

1. **S. Derhami** and A. E. Smith, [An integer programming approach for fuzzy rule-based classification systems](#), *European Journal of Operational Research*, vol. 256, no. 3, pp. 924–934, 2017.
2. **S. Derhami**, J. S. Smith, and K. R. Gue, [Optimising space utilisation in block stacking warehouses](#), *International Journal of Production Research*, vol. 55, no. 21, pp. 6436–6452, 2017.
3. **S. Derhami** and A. E. Smith, [A technical note on the paper “hGA: Hybrid genetic algorithm in fuzzy rule-based classification systems for high-dimensional problems”](#), *Applied Soft Computing*, vol. 41, pp. 91–93, 2016.
4. **S. Derhami**, J. S. Smith, and K. R. Gue, Space-efficient layouts for block stacking warehouses, *Submitted*.

### Refereed conference proceedings

1. **S. Derhami** and A. E. Smith, [Iterative mixed integer programming model for fuzzy rule-based classification systems](#), in *2014 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2014, pp. 2079–2084.
2. **S. Derhami**, J. S. Smith, and K. R. Gue, [A simulation model to evaluate the layout for block stacking warehouses](#), *To appear in Progress in Material Handling Research: 2016*.

### Working papers

1. **S. Derhami**, J. S. Smith, and K. R. Gue, Designing a robust layout for block stacking warehouses.
2. **S. Derhami**, B. Montreuil, and M. Ansari, Estimating consumer demand from sale transactions in a network of hyperconnected retailers under product substitution.

## Teaching experience

### Instructor

- Auburn University (Fall 2015)  
Stochastic Operations Research (98 students, avg. review: 5.0/6.0)
- Business Trading University, Iran (2007)  
Introduction to Manufacturing Processes, Production Management
- Azad University, Iran (2004-2005)  
Plant Layout, Inventory Management, Production Planning, Application of Computer in IE

### Teaching assistant

Operations Planning & Control, Stochastic Operation Research, Simulation (graduate), Probability & Statistics I, Simulation Modeling & Analysis, Deterministic Operations Research

## Honors & awards

- Finalist in the INFORMS data mining best student paper for the paper “An integer programming approach for fuzzy rule-based classification systems”, 2016
- Honorable award for the poster “A novel optimization approach for data mining: a medical diagnosis case”, Graduate Engineering Research Showcase, Auburn University, 2015 (\$1,000)
- Conveyor & Sortation Systems Honor Scholarship, the Material Handling Education Foundation, Inc., 2015 (\$2,000)
- Departmental award for the poster “Optimizing Space Utilization in Block Stacking Warehouses”, Graduate Engineering Research Showcase, Auburn University, 2014 (\$500)
- Storage equipment manufacturers association honor scholarship, the Material Handling Education Foundation, Inc., 2012 (\$3,000)
- Auburn University graduate travel grants to attend three conferences, 2014-2016 (\$2800)
- Graduate teaching assistantship, Auburn University, 2011-2017
- Graduate teaching assistantship, Bu-Ali Sina University, 2005-2006

## Conference presentations

- “Designing a space-efficient warehouse layout”, INFORMS, Nashville, TN, 2016
- “Optimizing space utilization in block stacking warehouses”, INFORMS, Philadelphia, PA, 2015
- “Extract fuzzy rules by integer programming”, IIE Annual Conference, Nashville, TN, 2015
- “Ant colony optimization to smooth work load in assembly lines”, 6th International Industrial Engineering Conference, February 18-19, Tehran, Iran, 2009
- “Using ant colony to solve multi-objective assembly line balancing problem”, 2nd International Operation Research Conference, May 20-22, Babolsar, Iran, 2009

## Service to profession

- Reviewer for European Journal of Operational Research, International Journal of Production Research, Computers & Operations Research, IEEE Transactions on Automation Science and Engineering, Applied Mathematical Modelling, SIMULATION: Transactions of The Society for Modeling and Simulation International, and 2015 IIE Annual Conference
- Session chair and student volunteer in 2015 IIE Annual Conference
- Member of Institute for Operations Research and the Management Sciences (INFORMS)
- Member of Institute of Industrial & Systems Engineers (IISE)

## References

- Dr. Jeffrey Smith  
Joe W. Forehand Jr. Professor  
Department of Industrial and Systems Engineering  
Auburn University  
Email: jsmith@auburn.edu
- Dr. Kevin Gue  
Professor and Mary Lee and George F. Duthie Endowed Chair of Engineering Logistics  
Department of Industrial Engineering  
University of Louisville  
Email: kevin.gue@louisville.edu
- Dr. Benoit Montreuil  
Professor and director of Supply Chain & Logistics Institute, and Physical Internet Center  
School of Industrial and Systems Engineering  
Georgia Institute of Technology  
Email: benoit.montreuil@isye.gatech.edu
- Dr. Alice Smith  
Joe W. Forehand/Accenture Distinguished Professor  
Department of Industrial and Systems Engineering  
Auburn University  
Email: smithae@auburn.edu